

LECTURE 29: MATH1061/7861
TOPICS: DEFINITIONS AND EXAMPLES OF
FIELDS
DATE: WEDNESDAY MAY 6, 2026

Learning objectives:

(1) Learn the definition of a field.

(2) Learn how to perform calculations and solve equations in fields.

$(F, +, \times)$ a field.

$\Rightarrow (F, +)$ group

$\{ \cancel{F} (F - \{0\}, \times) \text{ a group} \}$

* Final exam schedule has been released.

* Previous years' final exams & solutions are provided on the blackboard.

* Further information will be given in the last lecture.

Q1: Which of the following are fields?

(a) $(\mathbb{Z}, +, \times)$ *NOT a field.*

$(\mathbb{Z}, +)$ $\leftarrow$ group $\checkmark$

Is a field. $(\mathbb{Z} - \{0\}, \times)$ $\leftarrow$ Not closed under inverses.
 $\text{inv}(2) = \frac{1}{2} \notin \mathbb{Z}.$

(b) $(\mathbb{Z}_7, +, \times)$ $(\mathbb{Z}_7, +)$ is group $\checkmark$

$(\mathbb{Z}_7 - \{0\}, \times)$ saw last time this is a group.
(c) $(\mathbb{Z}_{15}, +, \times)$

$\hookrightarrow$ Not a field.

$$[3] \cdot [5] = [0]$$

So $\mathbb{Z}_{15} - \{0\}$ is not closed
under multiplication.

In a similar way one shows that if
 n is composite then $\mathbb{Z}_n$ is not a field.

Note: If n is prime then $\mathbb{Z}_n$ is a field.
(But this is not obvious²)

In $\mathbb{Z}$, we have distributive:

$$a, b, c \in \mathbb{Z},$$

$$a(b+c) = ab+ac$$

In $\mathbb{Z}_n$, we claim:

$$[a] \cdot ([b] + [c]) = [ab] + [ac]$$

The reason this works is because of the way addition & multiplication of these equivalence classes are defined:

Namely:

$$\left. \begin{aligned} [a] + [b] &= [a+b] \\ [a] \times [b] &= [a \times b] \end{aligned} \right\}$$

Q2: Determine the additive and multiplicative inverse of 7 in $(\mathbb{Z}_{100}, +, \times)$.

↑
Not a field.

Additive inverse: $-[7] = [-7] = [43]$

Multiplicative inverse: $[7]^{-1} = \boxed{x} \boxed{y} = \underline{\underline{43}}$
 ↖ ↗
 digits.

$$7 \times \boxed{x} \boxed{y} = \text{multiple of } 100 + 1$$

last digit of this has to be 1.

$$7 \times 3 = 21$$

$$7 \times 13 = 91$$

$$7 \times 23$$

$$7 \times 33$$

$$7 \times 43 = 301$$

Q3: Solve $3x + 1 = 5$ in the field $(\mathbb{Z}_7, +, \times)$.

$$3x + \cancel{1} = 5$$

$$\Rightarrow 3x = 4$$

$$\Rightarrow x = 3^{-1} \times 4$$

$$\Rightarrow x = 5 \times 4$$

$$\Rightarrow \underline{\underline{x = 20 = 6}}$$

Q4: Solve the equation $2x + 3 = 8$ in the field $(\mathbb{Z}_{11}, +, \times)$.

$$2x + 3 = 8$$

$$\Rightarrow 2x = 8 - 3 = 5$$

$$\Rightarrow x = (2^{-1}) \times 5$$

$$\Rightarrow x = 6 \times 5 = 30 = 8$$

$$\Rightarrow \boxed{x = 8}$$